Functional programming and formal linguistics

HHU Winter Semester 2025

Patrick D. Elliott

October 13, 2025

1 Description

The philosopher Richard Montague began his 1970 paper "Universal Grammar" with the following (at the time, controversial) claim:

> "There is in my opinion no important theoretical difference between natural languages and the artificial languages of logicians; indeed, I consider it possible to comprehend the syntax and semantics of both kinds of language within a single natural and mathematically precise theory."

In this class, we'll update Montague's claim for the 21st century: there's no important difference between natural languages and the programming languages of computer scientists. Concretely, we'll explore connections between the formal tools used for modelling the syntax and semantics of natural languages and the so-called 'functional' programming paradigm. This will be a practical class in which we'll first develop a working knowledge of the functional programming language Haskell, drawing from recent popular textbooks and online resources. We'll use Haskell to implement simple algorithms taken from linguistic theory, focusing on logic and semantics. Material will be drawn from van Eijck and Unger's 2012 textbook "Computational Semantics with Functional Programming".

2 Team

- **Instructor:** Dr. Patrick D. Elliott
  - Patrick.Elliott@hhu.de
  - `https://patrickdelliott.com`
- **Secretary:** Tim Martion

– tim.marton@phil.hhu.de

3 Practical information

3.1 Pre-requisites

- Introduction to semantics (or equivalent).
- Some basic programming experience is desirable but not necessary.

3.2 Laptops

If possible, please bring a laptop with you to class.

3.3 Time and place

- Class takes place on **Tuesday** at 16:30-18:00.
- Class takes place in **2321.04.23 (Z 34)**.

3.4 Language of instruction

The lectures, as well as the readings for this class will be in **English**.

4 Leistungsnachweis

4.1 BN requirements

- Participate in class.
- Submit solutions to the occasional homework assignments.

4.2 AP requirements

- Fulfill the BN requirements.
- Complete a final project consisting of:
  - A written technical report/a simple haskell program with documentation.
  - An accompanying in-class presentation.

5 Comms

5.1 Rocketchat

Please join the rocketchat channel using the provided link. Note that this is **obligatory**; in an attempt to keep things simple, I'll use this as the main channel of communication for the class, including announcements etc.

5.2 Office hours

I hold office hours at 11-12pm on Wednesdays. You can find me in building 23.21, room 04.73. Please let me know in advance the time that you plan on dropping by, so that I can stagger appointments.

5.3 Email

For any questions regarding the class, please post in rocketchat, so that others can benefit from the response. For any private queries, you can email me at Patrick.Elliott@hhu.de.

6 Readings

6.1 Main readings

- *Haskell from first principles* (FP) (Allen & Moronuki 2020)
- *Computational semantics with functional programming* (CS) (Van Eijck & Unger 2010)

6.2 Supplementary

- *The haskell road to logic, maths, and programming* (LMP) (Doets & van Eijck 2012)

7 Resources for getting started with haskell

- The haskell playground

`https://play.haskell.org/`

8 Tentative schedule

date	class
October 14	Lambda calculus
October 21	Intro to Haskell
October 28	Strings
November 4	Basic datatypes
November 11	Types and typeclasses
November 18	NO CLASS
November 25	Fragments i: syntax
December 2	Fragments ii: semantics
December 9	Model checking
December 16	Meaning composition
January 6	Functors
January 13	Applicatives
January 20	Monads
January 27	Monads cont.
February 3	Student presentations

References

Allen, Christopher & Julie Moronuki. 2020. *Haskell programming from first principles*. Lorepub LLC.

Doets, Kees & Jan van Eijck. 2012. *The Haskell road to logic, maths and programming*. 2. ed (Texts in Computing 4). London: College Publ. 432 pp.

Van Eijck, Jan & Christina Unger. 2010. *Computational Semantics with Functional Programming*. 1st edn. Cambridge University Press. `https://www.cambridge.org/core/product/identifier/9780511778377/type/book` (4 April, 2024).